THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET A

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Matter is made up of: 1

(a) only liquids **(b)** only atoms of gold
(c) extremely small particles **(d)** continuous material without gaps

2. Interparticle attraction is weakest in: 1

(a) liquids **(b)** solids **(c)** ice **(d)** gases

3. Gases have: 1

(a) no fixed shape and no fixed volume **(b)** fixed shape
(c) fixed volume **(d)** fixed density

4. Gas in a syringe can be compressed because: 1

(a) particles are touching **(b)** gas is solid
(c) large interparticle spaces exist **(d)** gas has fixed shape

5. The melting point of ice is: 1

(a) 1538 °C **(b)** 100 °C **(c)** 133 °C **(d)** 0 °C

6. When a solid is heated, its particles: 1

(a) stop moving **(b)** become bigger
(c) vibrate more vigorously **(d)** turn into atoms of another kind

7. Smell of food reaches you from the kitchen because particles of the smell: 1

(a) spread among air particles **(b)** are stationary
(c) stay in kitchen **(d)** disappear

8. Salt in sea water is not visible because: 1

(a) salt turns into water
(b) salt turns into gas
(c) salt disappears forever
(d) its particles are mixed in very small sizes among water particles

9. Rice flour poured into a jar takes the shape of the jar, but it is still: 1

(a) a liquid **(b)** a gas
(c) plasma **(d)** a solid made of tiny solid grains

10. The state of wax in a melted candle near the flame is: 1

(a) gas only **(b)** plasma **(c)** solid only **(d)** liquid

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is meant by the particulate nature of matter? 2

12. What are interparticle spaces? In which state are they minimum? 2

13. State two properties of solids based on their particles. 2

14. Why can liquids flow but solids cannot? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Compare solids, liquids and gases on shape, volume and interparticle space. 3

16. Explain the movement of particles in solids, liquids and gases. 3

17. Describe the changes in particle arrangement when ice melts into water and then into vapour. 3

OR

Explain the term melting point with the help of particle behaviour. 3

18. Explain boiling point with particle behaviour. 3

OR

Describe the syringe activity with air and water and give conclusions. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain the particulate nature of matter. Describe the arrangement, movement and forces of particles in the three states. 5

OR

Compare solids, liquids and gases in a table (shape, volume, compressibility, flow, interparticle space). 5

20. Describe melting and boiling in terms of particles. Define melting point and boiling point. 5

OR

Explain with activities that gases are compressible, but liquids and solids are not. 5

END OF QUESTION PAPER — Total: 40 Marks

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